Exit ticket · Lesson 4.1 Water · Work alone · 3 minutes · Name: _____

1. Recall. What is surface tension?

2. Understanding. Explain why ice floating on a lake helps fish survive the winter.

3. Application. A tall tree needs water in its top leaves. Use cohesion and adhesion to explain how the water gets there.

✂ - - -

Name / Имя: _____ **Class:** _____ **Date:** _____

Laboratory Work No. 2 · Drops on a coin

Aim: observe how water molecules stick together (cohesion) and stick to the coin (adhesion).

Materials	Safety
• a glass of water • a dropper (pipette) • a coin • a paper towel	• Keep water away from sockets and devices. • Wipe spills at once. • Wash your hands after the lab.

1 · Pre-lab questions · Вопросы до работы

a) What is cohesion?

b) What other properties does water have?

2 · Predict · Предскажи

I predict _____ drops will stay on the coin before the water spills, because _____

3 · Procedure · Ход работы

1. Put the coin on a flat surface. Dry it with the towel.
2. Fill the dropper with water.
3. Slowly drip water onto the coin — **one drop at a time**.
4. Count the drops. Write the number when the coin is **full**.
5. Keep dripping. Write the number when the water **spills**.
6. Dry the coin. Repeat (trial 2, trial 3). Swap roles.

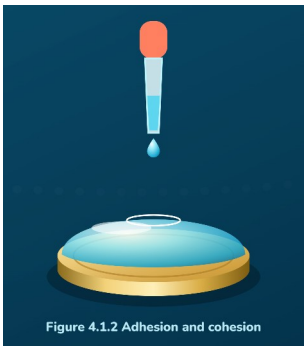


Figure 4.1.2 Adhesion and cohesion

4 · Results · Результаты

Write the **real** number of drops you counted.

Trial	Drops when the coin is full	Drops when the water spills
1		
2		
3		
Mean (average)		

Class results (from the board): lowest _____ highest _____ mean _____

Draw the water on the coin from the side. Label: coin, water, surface.

5 · Post-lab questions · Вопросы после работы

1. Why did the water on the coin not spill?

The water did not spill because _____

2. In which processes is this property used? How does it help living organisms?

3. What would happen if water molecules could not stick to each other?

If water molecules could not stick together, then _____

4. Why were the numbers of different pairs not the same? Give one reason.

Name / Имя: _____ **Class:** _____ **Date:** _____

Vocabulary Словарь · Вода

Read the word. Look at the picture. Say it in English. Keep this sheet on your desk during the lesson.

	English	Русский	Meaning (simple English)
	**solvent**	растворитель	a liquid in which solids dissolve
	**dissolve**	растворяться	a solid mixes into a liquid and we cannot see it
	**surface tension**	поверхностное натяжение	a property of the surface of a liquid that allows it to resist an external force
	**cohesion**	когезия (сцепление)	water molecules stick to each other
	**adhesion**	адгезия (прилипание)	water molecules stick to other molecules
	**molecule**	молекула	a very small particle; a water molecule is H_2O
	**specific heat capacity**	удельная теплоёмкость	the energy needed to raise the temperature of 1 g of a substance by 1 °C
	**density**	плотность	how much mass is packed into a space
	**melting**	плавление	solid ice turns into liquid water (0 °C)
	**boiling**	кипение	liquid water turns into water vapour (100 °C)
	**water vapour**	водяной пар	water as a gas
	**states of water**	состояния воды	solid (ice), liquid (water), gas (vapour)

Practice 1 · Fill the gaps · Заполни пропуски

Word bank: solvent · cohesion · adhesion · melts · boils · floats

1. Water is a _____: many solids dissolve in it.
2. Water molecules stick to each other. This is _____.
3. Water sticks to the glass. This is _____.
4. Ice _____ at 0 °C.
5. Water _____ at 100 °C.
6. Ice is less dense than water, so it _____.

Practice 2 · English → Русский · Переведи

English	Русский (write)
solvent	
cohesion	
specific heat capacity	
boiling	

Practice 3 · My sentence · Моё предложение

Write one sentence. Use **two** words from this sheet.

Exit ticket · Lesson 4.2 Chemical elements in human body · Work alone · 3 minutes · Name: _____

1. Recall. Name the four main elements that make up 96% of the body.

2. Understanding. What is the difference between a macroelement and a microelement? Give one example of each.

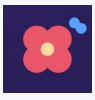
3. Application. A person's muscles do not contract well and their blood does not clot normally. Which element could be missing? Explain.

✂ - - -

Name / Имя: _____ **Class:** _____ **Date:** _____

Vocabulary Словарь · Химические элементы в организме человека

Read the word. Look at the picture. Say it in English. Keep this sheet on your desk during the lesson.

	English	Русский	Meaning (simple English)
	**element**	элемент	a substance that cannot be divided into other substances by chemical reactions
	**bioelements**	биоэлементы	the units which living things are made of
	**chemical symbol**	химический символ	1–2 letters for an element; Na comes from Latin natrium
	**macroelement**	макроэлемент	an element needed in large amounts
	**microelement**	микроэлемент	an element needed in small (trace) amounts
	**deficiency**	недостаток (дефицит)	not enough of something the body needs
	**calcium (Ca)**	кальций	in bones and teeth; muscle contraction; blood clotting
	**phosphorus (P)**	фосфор	in bones, teeth and nucleic acids; energy transmission
	**potassium (K)**	калий	main role in the nervous system
	**iron (Fe)**	железо	part of hemoglobin in the blood
	**iodine (I)**	йод	main component of the thyroid hormones
	**hemoglobin**	гемоглобин	a protein in the blood that transports oxygen

Practice 1 · Fill the gaps · Заполни пропуски

Word bank: hemoglobin · macroelement · microelement · teeth · deficiency · thyroid

1. Iron is a part of _____, which carries oxygen in the blood.
2. An element that the body needs in large amounts is a _____.
3. Iodine is needed in very small amounts. It is a _____.
4. Calcium and phosphorus are found in bones and _____.
5. Not enough of an element is called a _____.
6. Iodine is the main component of the _____ hormones.

Practice 2 · English → Русский · Переведи

English	Русский (write)
element	
macroelement	
calcium (Ca)	
iron (Fe)	

Practice 3 · My sentence · Моё предложение

Write one sentence. Use **two** words from this sheet.

Name / Имя: _____ Class: _____ Date: _____

Which part of the body will be affected?

Textbook Activity p. 59. Work in pairs. Use the information box.

Information box · Информация

Element	What it does in the body (textbook pp. 58–59)
Calcium (Ca)	Found in bones and teeth. Acts in muscle contraction and blood clotting.
Phosphorus (P)	Found in bones and teeth and in nucleic acids. Helps in energy transmission.
Potassium (K)	Main role in the nervous system. Na, K and Cl form the nerve impulses.
Magnesium (Mg)	Role in protein production and relaxation of muscles.
Iron (Fe)	Part of hemoglobin — the blood protein that transports oxygen.
Iodine (I)	Main component of the thyroid hormones.
Fluorine (F)	A microelement. Teacher information: it helps keep teeth strong (not explained in the textbook).

Task A · Complete the table · Заполни таблицу

Word bank: **thyroid gland · blood · bones · teeth · muscles · nervous system**

Element	Macro or micro?	Part of the body affected	Because… (reason)
Iodine (I)			
Fluorine (F)			
Iron (Fe)			
Calcium (Ca)			
Phosphorus (P)			
Potassium (K)			

Example: If the body lacks iodine, the thyroid gland will be affected because iodine is the main part of thyroid hormones.

Task B · Read the chart (Figure 4.2.1) · Прочитай диаграмму

Element	O	C	H	N	Ca	P	K	S	Na	Cl	Mg
% of body	65	18.5	9.5	3.3	1.5	1	0.4	0.3	0.2	0.2	0.1

Microelements together make up less than 0.1%.

1. Which element is the most abundant in the body?

2. Add the percentages of O, C, H and N. Show your working. What do you notice?

3. Iron is less than 0.1% of the body. Is it still important? Explain.

Task C · Element detective · Детектив

A 12-year-old is growing fast. Which **two** elements help to build bones and teeth? Use the information box.

> I think _____ and _____ are needed because _____

Challenge: Nerve signals are weak. Which elements form nerve impulses?

Exit ticket · Lesson 4.3 Organic compounds · Work alone · 3 minutes · Name: _____

1. Recall. Which reagent tests for protein, and what colour change shows protein?

2. Understanding. Why does an olive leave a transparent spot on paper, but a potato does not?

3. Application. A student adds iodine solution to bread. It turns blue-black. What does this show? Which group is bread in?

Exit ticket · Lesson 4.3 Organic compounds · Work alone · 3 minutes · Name: _____

1. Recall. Which reagent tests for protein, and what colour change shows protein?

2. Understanding. Why does an olive leave a transparent spot on paper, but a potato does not?

3. Application. A student adds iodine solution to bread. It turns blue-black. What does this show? Which group is bread in?

Name / Имя: _____ Class: _____ Date: _____

Laboratory Work No. 3 · Testing for organic molecules

Aim: find out which foods contain carbohydrates (starch), lipids and proteins.

Safety check · tick each box · Техника безопасности

✓	Rule
☐	I wear gloves and eye protection when reagents are used.
☐	I never taste the food samples.
☐	Only the teacher pours iodine solution and Biuret reagent.
☐	I tell the teacher about spills at once.
☐	I wash my hands after the lab.
☐	I told the teacher if I am allergic to egg or olives.

1 · Pre-lab questions · Вопросы до работы

a) Why do we need carbohydrates?

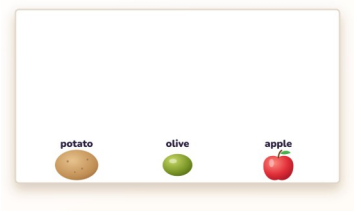
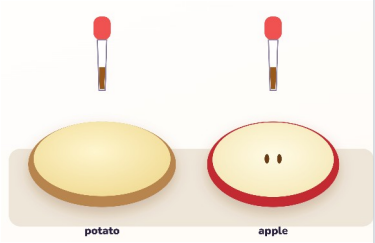
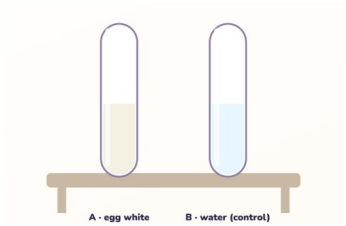
b) What is the role of proteins?

c) What is the importance of fats?

2 · Predict · Предскажи

Food	I think it contains… (carbohydrate / lipid / protein)	Because…
potato		
apple		
olive		
egg white		

3 · Procedure · Ход работы

Test	What to do	Result rule
A · Paper test (lipids) 	1. Squeeze and knead potato, olive and apple separately on white paper until wet. 2. Clean off the food. Let the paper dry for 10–15 minutes. 3. Hold the paper up to the light. Compare the traces.	If the food contains fat, the trace is **transparent**.
B · Iodine test (carbohydrates) 	1. The teacher cuts the potato and apple in half. 2. Add a few drops of iodine solution to each. 3. Write the colour you see.	With carbohydrate (starch), colour changes from **brown to blue or black**.
C · Biuret test (proteins) 	1. Put 2 ml raw egg white in a test tube. 2. The teacher adds 2 ml Biuret reagent. Mix. 3. Write the result. Compare with water.	With protein, colour changes from **blue to purple**.

Start test A first — the paper needs time to dry.

4 · Observations · Наблюдения

Write what **you** saw. Do not copy from the board.

Test	Food	Colour / trace I saw	Compound present? (yes / no)
Iodine	potato		
Iodine	apple		
Paper	potato		
Paper	olive		
Paper	apple		
Biuret	egg white		
Biuret	water (control)		

5 · Explain · Объясни

I observed _____. This suggests _____ because _____.

6 · Post-lab questions · Вопросы после работы

1. What foods contain carbohydrates?

2. What foods are rich in protein?

3. From what foods do we get fat?

4. Which result was hard to see? How could we make the test fairer?

Name / Имя: _____ **Class:** _____ **Date:** _____

Vocabulary Словарь · Органические соединения

Read the word. Look at the picture. Say it in English. Keep this sheet on your desk during the lesson.

	English	Русский	Meaning (simple English)
	**organic compound**	органическое соединение	a compound that contains carbon and hydrogen
	**carbohydrate**	углевод	organic compound with C, H, O (sugars); main source of energy
	**protein**	белок	basic building organic compound of the body (C, H, O, N, S)
	**lipid (fat)**	липид (жир)	stores the most energy; insoluble in water
	**energy**	энергия	what living things need to move, grow and work
	**insoluble**	нерастворимый	does not dissolve in water
	**iodine solution**	йодный раствор	brown liquid; turns blue-black with starch
	**Biuret reagent**	биуретовый реактив	blue liquid; turns purple with protein
	**test tube**	пробирка	a small glass tube for experiments
	**pipette**	пипетка	a tool that adds a few drops
	**transparent spot**	прозрачное пятно	a mark on paper that lets light through
	**colour change**	изменение цвета	the colour becomes different — a sign of a reaction

Practice 1 · Fill the gaps · Заполни пропуски

Word bank: protein · starch · transparent · insoluble · energy · carbon

1. Biuret reagent turns purple when the food contains _____.
2. Iodine solution turns blue-black with _____ (a carbohydrate).
3. Oil leaves a _____ spot on paper.
4. Lipids are _____ in water.
5. Carbohydrates are the main source of _____.
6. Organic compounds contain _____ and hydrogen.

Practice 2 · English → Русский · Переведи

English	Русский (write)
organic compound	
lipid (fat)	
iodine solution	
pipette	

Practice 3 · My sentence · Моё предложение

Write one sentence. Use **two** words from this sheet.

Exit ticket · Lesson 4.4 Deficiency of macroelements in plants · Work alone · 3 minutes · Name: _____

1. Recall. Name two organic fertilizers.

2. Understanding. Why is watering not enough to grow a healthy plant?

3. Application. A plant's leaf edges turn yellow, then brown, and the plant is damaged by frost. Which element is missing? Explain.

✂ - - -

Name / Имя: _____ **Class:** _____ **Date:** _____

Vocabulary Словарь · Недостаток макроэлементов у растений

Read the word. Look at the picture. Say it in English. Keep this sheet on your desk during the lesson.

	English	Русский	Meaning (simple English)
	**deficiency**	недостаток	not enough of a needed element
	**fertilizer**	удобрение	a natural or artificial substance added to soil to help plants grow better
	**macroelement**	макроэлемент	an element needed in large amounts (e.g. N, P, K)
	**organic fertilizer**	органическое удобрение	natural: from plant or animal products (manure, compost)
	**inorganic fertilizer**	неорганическое удобрение	synthetic: made in a factory
	**synthetic**	синтетический	made artificially by combining separate elements
	**vein**	жилка	a thin line in a leaf that carries water
	**edge (margin)**	край листа	the outside line of a leaf
	**tip**	кончик листа	the end of the leaf
	**pale / dark**	бледный / тёмный	light colour / deep colour
	**spots**	пятна	small marks of a different colour
	**ripen**	созревать	a fruit becomes ready to eat

Practice 1 · Fill the gaps · Заполни пропуски

Word bank: organic · inorganic · veins · deficiency · edge · ripen

1. Manure and compost are _____ fertilizers.
2. Superphosphate is made in a factory. It is an _____ fertilizer.
3. The thin lines in a leaf are called _____.
4. When a plant does not get enough nitrogen, it has a nitrogen _____.
5. The outside line of a leaf is the _____.
6. Without enough phosphorus, fruits do not _____.

Practice 2 · English → Русский · Переведи

English	Русский (write)
deficiency	
organic fertilizer	
vein	
pale / dark	

Practice 3 · My sentence · Моё предложение

Write one sentence. Use **two** words from this sheet.

Name / Имя: _____ **Class:** _____ **Date:** _____

Leaf detectives · Match the description

Textbook Activity p. 63. Work in pairs. Read each description. Underline the key words.

Figure 4.4.1 · Nutrient deficiency in plants · Недостаток питательных веществ



healthy	nitrogen (N)	phosphorus (P)	potassium (K)	magnesium (Mg)

Task A · Match · Сопоставь

Choose from: **A** Phosphorus deficiency **B** Magnesium deficiency **C** Nitrogen deficiency **D** Potassium deficiency

Description	Letter	Key words that helped me
1. The old leaves turn brownish-yellow and gradually disappear. Yellowing of the leaves begins with the leaf veins.		
2. The colour of the old leaves turns dark green. Brown and pink spots appear on the leaves and then turn into holes.		
3. The leaves turn yellow and later dry out at the tips and edges. Brown spots appear on the edges.		
4. Pale yellow spots appear between the veins. Large nerves (veins) and leaf margins are green. The edge of the leaf is bent and the leaf shape is domed.		

I chose ___ for description ___ because it says “_____”.

Task B · Organic or inorganic? · Органическое или неорганическое?

Fertilizer	Organic or inorganic?
manure	
superphosphate	
compost	
potassium chloride	
guano	
ammonium sulfate	

Task C · Plant doctor · Доктор растений

A tomato plant has few, small leaves, and its fruits do not ripen.

a) Which element is missing?

b) Which fertilizer could help? Why?

Task D · Think about it · Подумай

Is “more fertilizer” always better for plants? Give one reason.

> I think _____ because _____